

Lab 11 Activity

We will work with the same dataset that we used in the main lab. You can read in the data using the following code:

```
library(tidyverse)
library(haven)
library(lavaan)
dat <- read_dta("Lab_11_attend.dta")
```

1

The variables `frosh` and `soph` are dummy variables for the student being a freshman or sophomore (0 on both variables indicates 'other'). Fit a model predicting final exam scores from (the centered) prior GPA with academic year as a moderator (no need to modify the categorical moderator variables). Is there a statistically significant moderation effect?

```
dat <- dat %>% mutate(priGPA_C = priGPA - mean(priGPA),
                     atndrte_C = atndrte - mean(atndrte))
mod <- lm(final ~ priGPA_C * frosh + priGPA_C * soph, dat)
mod <- lm(final ~ priGPA_C * (frosh + soph), dat) # another way to specify the same model
summary(mod)
```

Call:

```
lm(formula = final ~ priGPA_C * (frosh + soph), data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-16.5846	-2.8601	-0.0644	2.8344	11.6477

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	26.3465	0.3829	68.810	< 2e-16 ***
priGPA_C	4.1172	0.7858	5.239	2.16e-07 ***
frosh	-0.8439	0.5626	-1.500	0.13408
soph	-0.7571	0.4440	-1.705	0.08857 .
priGPA_C:frosh	-2.9060	1.0735	-2.707	0.00696 **
priGPA_C:soph	-0.4441	0.8843	-0.502	0.61567

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.356 on 674 degrees of freedom

Multiple R-squared: 0.1508, Adjusted R-squared: 0.1445

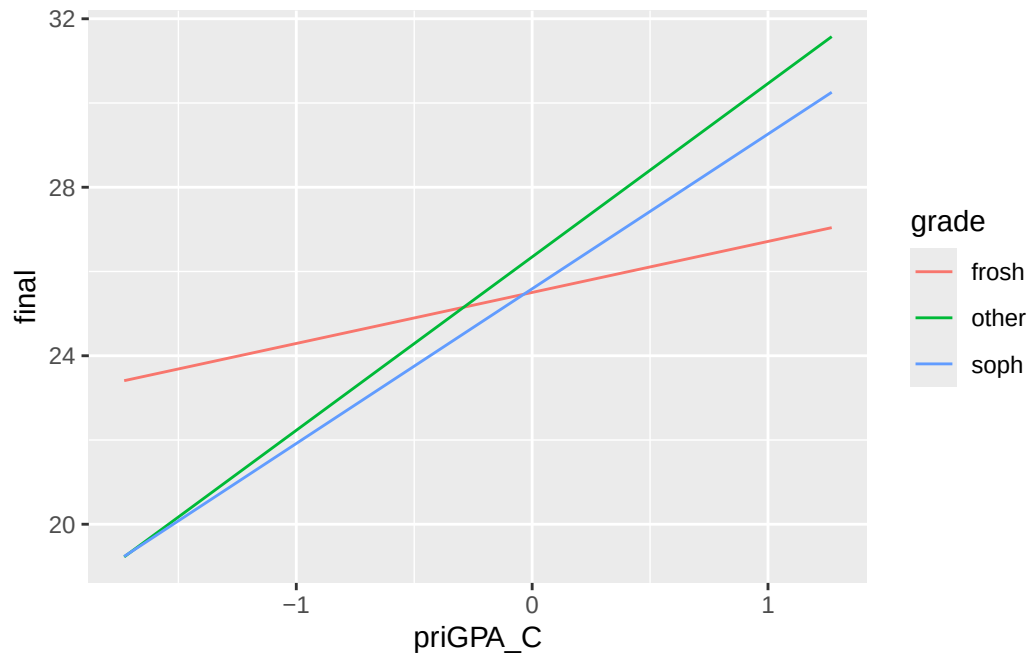
F-statistic: 23.93 on 5 and 674 DF, p-value: < 2.2e-16

the moderation effect is significant to freshman, but not for sophomores

2

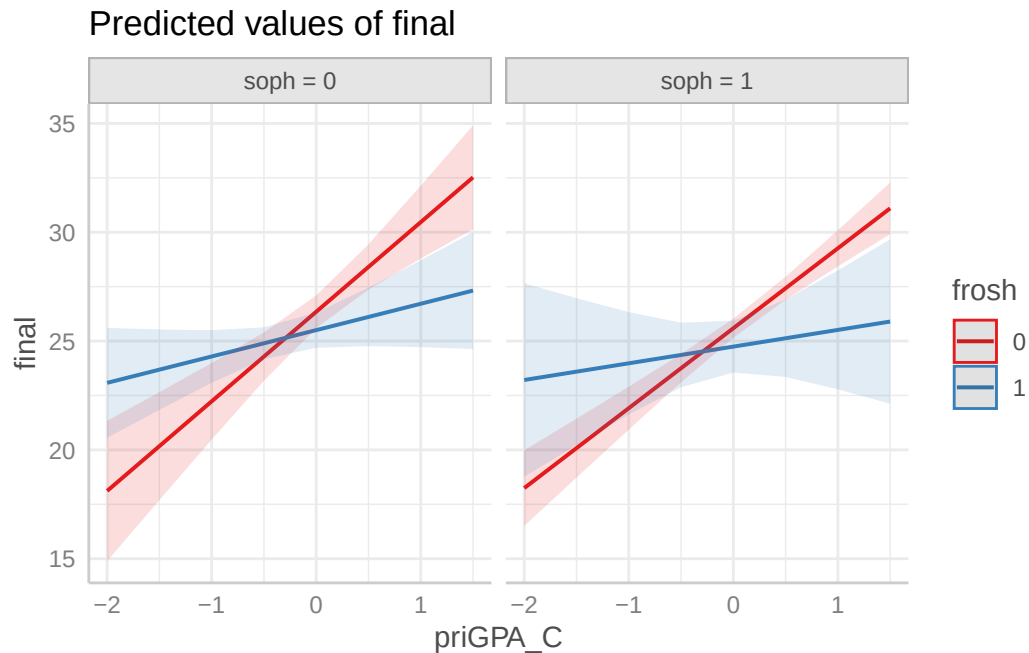
Construct a plot to illustrate the moderation effect (use separate lines for different grade levels). How does this plot help you interpret your results from step 1?

```
dat_plot <- tibble(priGPA_C = seq(min(dat$priGPA_C), max(dat$priGPA_C)))
dat_plot <- dat_plot %>%
  mutate(frosh = 26.3465 - 0.8439 + priGPA_C * (4.1172 - 2.9060),
         soph = 26.3465 - 0.7571 + priGPA_C * (4.1172 - 0.4441),
         other = 26.3465 + 4.1172 * priGPA_C) %>%
  pivot_longer(-priGPA_C, names_to = "grade", values_to = "final")
ggplot(dat_plot, aes(priGPA_C, final, col = grade)) + geom_path()
```



```
## There is a positive (predicted) relationship between prior GPA and  
## final for all grade levels, but this relationship has a less steep  
## slope for freshmen
```

```
## alternative:  
library(ggeffects)  
ggp <- ggpredict(mod, terms = c("priGPA_C", "frosh", "soph"))  
plot(ggp)
```



3

Using this same data set, conduct a mediation analysis predicting final exam scores from ACT scores with `hwrtte` (rate of completing homework) as a mediator (you may drop the 6 missing values of `hwrtte` for this step). What do you conclude?

```
mod_m <- lm(hwrtte ~ ACT, dat)
mod_y <- lm(final ~ ACT + hwrtte, dat)

library(mediation)
```

Loading required package: MASS

Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

`select`

Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

expand, pack, unpack

Loading required package: mvtnorm

Loading required package: sandwich

mediation: Causal Mediation Analysis

Version: 4.5.1

```
summary(mediate(model.m = mod_m, model.y = mod_y,  
                treat = "ACT", mediator = "hwrt", boot = TRUE))
```

Running nonparametric bootstrap

Causal Mediation Analysis

Nonparametric Bootstrap Confidence Intervals with the Percentile Method

	Estimate	95% CI Lower	95% CI Upper	p-value
ACME	-0.021593	-0.045037	-0.001682	0.036 *
ADE	0.512496	0.417550	0.609747	<2e-16 ***
Total Effect	0.490903	0.395166	0.589393	<2e-16 ***
Prop. Mediated	-0.043987	-0.096818	-0.003419	0.036 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Sample Size Used: 674

Simulations: 1000

```
## The mediation effect is statistically
## significant, although the effect is rather small
## (especially compared to the mediator term GPA that we used
## in lab). Using the effect size measures from the main lecture
## could help clarify whether this effect is practically significant
```

4

Repeat #3 using a method that using a better method of missing data handling than listwise deletion.

```
medmod <- "
final ~ b * hwrt + c * ACT
hwrt ~ a * ACT

ind := a*b    # calculate the indirect effect
tot := a*b+c  # calculate the direct effect
"

fitted <- sem(medmod, data = dat, se = "boot",
             missing = "FIML", estimator = "ML") #, estimator = "MLR") # robust standard errors

summary(fitted)
```

lavaan 0.6-20 ended normally after 23 iterations

Estimator	ML
Optimization method	NLMINB
Number of model parameters	7
Number of observations	680
Number of missing patterns	2

Model Test User Model:

Test statistic	0.000
Degrees of freedom	0

Parameter Estimates:

Standard errors	Bootstrap
-----------------	-----------

Number of requested bootstrap draws	1000
Number of successful bootstrap draws	1000

Regressions:

		Estimate	Std.Err	z-value	P(> z)
final ~					
hwrt	(b)	0.044	0.009	4.931	0.000
ACT	(c)	0.509	0.048	10.546	0.000
hwrt ~					
ACT	(a)	-0.492	0.216	-2.281	0.023

Intercepts:

	Estimate	Std.Err	z-value	P(> z)
.final	10.577	1.393	7.596	0.000
.hwrt	98.961	4.782	20.693	0.000

Variances:

	Estimate	Std.Err	z-value	P(> z)
.final	18.551	0.929	19.965	0.000
.hwrt	367.774	32.891	11.181	0.000

Defined Parameters:

	Estimate	Std.Err	z-value	P(> z)
ind	-0.022	0.011	-2.054	0.040
tot	0.487	0.048	10.123	0.000

Defined Parameters:

#		Estimate	Std.Err	z-value	P(> z)
#	ind	0.149	0.027	5.447	0.000
#	tot	0.487	0.048	10.104	0.000

brms for the curious

```
library(mice)
```

Attaching package: 'mice'

The following object is masked from 'package:stats':

filter

The following objects are masked from 'package:base':

cbind, rbind

```
library(brms)
```

Loading required package: Rcpp

Loading 'brms' package (version 2.23.0). Useful instructions can be found by typing `help('brms')`. A more detailed introduction to the package is available through `vignette('brms_overview')`.

Attaching package: 'brms'

The following object is masked from 'package:stats':

ar

```
imputed_data <- mice(dat, m = 5)
```

```
iter imp variable
1    1  hwrte
1    2  hwrte
1    3  hwrte
1    4  hwrte
1    5  hwrte
2    1  hwrte
2    2  hwrte
2    3  hwrte
2    4  hwrte
2    5  hwrte
3    1  hwrte
3    2  hwrte
3    3  hwrte
3    4  hwrte
3    5  hwrte
4    1  hwrte
4    2  hwrte
```



```

4    3  hwrte
4    4  hwrte
4    5  hwrte
5    1  hwrte
5    2  hwrte
5    3  hwrte
5    4  hwrte
5    5  hwrte

```

Warning: Number of logged events: 5

```

brms_form <- bf(final ~ hwrte + ACT) + bf(hwrte ~ ACT) + set_rescor(FALSE)

brms_fitted <- brm_multiple(brms_form, data = imputed_data)

```

Compiling the C++ model

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).

Chain 1:

Chain 1: Gradient evaluation took 2.7e-05 seconds

Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 0.27 seconds.

Chain 1: Adjust your expectations accordingly!

Chain 1:

Chain 1:

Chain 1: Iteration: 1 / 2000 [0%] (Warmup)

Chain 1: Iteration: 200 / 2000 [10%] (Warmup)

Chain 1: Iteration: 400 / 2000 [20%] (Warmup)

Chain 1: Iteration: 600 / 2000 [30%] (Warmup)

Chain 1: Iteration: 800 / 2000 [40%] (Warmup)

Chain 1: Iteration: 1000 / 2000 [50%] (Warmup)

Chain 1: Iteration: 1001 / 2000 [50%] (Sampling)

Chain 1: Iteration: 1200 / 2000 [60%] (Sampling)

Chain 1: Iteration: 1400 / 2000 [70%] (Sampling)

Chain 1: Iteration: 1600 / 2000 [80%] (Sampling)

Chain 1: Iteration: 1800 / 2000 [90%] (Sampling)

Chain 1: Iteration: 2000 / 2000 [100%] (Sampling)

Chain 1:

Chain 1: Elapsed Time: 0.045 seconds (Warm-up)

Chain 1: 0.027 seconds (Sampling)

Chain 1: 0.072 seconds (Total)

Chain 1:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).

Chain 2:

Chain 2: Gradient evaluation took 4e-06 seconds

Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 0.04 seconds.

Chain 2: Adjust your expectations accordingly!

Chain 2:

Chain 2:

Chain 2: Iteration: 1 / 2000 [0%] (Warmup)

Chain 2: Iteration: 200 / 2000 [10%] (Warmup)

Chain 2: Iteration: 400 / 2000 [20%] (Warmup)

Chain 2: Iteration: 600 / 2000 [30%] (Warmup)

Chain 2: Iteration: 800 / 2000 [40%] (Warmup)

Chain 2: Iteration: 1000 / 2000 [50%] (Warmup)

Chain 2: Iteration: 1001 / 2000 [50%] (Sampling)

Chain 2: Iteration: 1200 / 2000 [60%] (Sampling)

Chain 2: Iteration: 1400 / 2000 [70%] (Sampling)

Chain 2: Iteration: 1600 / 2000 [80%] (Sampling)

Chain 2: Iteration: 1800 / 2000 [90%] (Sampling)

Chain 2: Iteration: 2000 / 2000 [100%] (Sampling)

Chain 2:

Chain 2: Elapsed Time: 0.061 seconds (Warm-up)

Chain 2: 0.028 seconds (Sampling)

Chain 2: 0.089 seconds (Total)

Chain 2:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).

Chain 3:

Chain 3: Gradient evaluation took 4e-06 seconds

Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 0.04 seconds.

Chain 3: Adjust your expectations accordingly!

Chain 3:

Chain 3:

Chain 3: Iteration: 1 / 2000 [0%] (Warmup)

Chain 3: Iteration: 200 / 2000 [10%] (Warmup)

Chain 3: Iteration: 400 / 2000 [20%] (Warmup)

Chain 3: Iteration: 600 / 2000 [30%] (Warmup)

Chain 3: Iteration: 800 / 2000 [40%] (Warmup)

Chain 3: Iteration: 1000 / 2000 [50%] (Warmup)

Chain 3: Iteration: 1001 / 2000 [50%] (Sampling)

Chain 3: Iteration: 1200 / 2000 [60%] (Sampling)

Chain 3: Iteration: 1400 / 2000 [70%] (Sampling)

Chain 3: Iteration: 1600 / 2000 [80%] (Sampling)
 Chain 3: Iteration: 1800 / 2000 [90%] (Sampling)
 Chain 3: Iteration: 2000 / 2000 [100%] (Sampling)
 Chain 3:
 Chain 3: Elapsed Time: 0.06 seconds (Warm-up)
 Chain 3: 0.026 seconds (Sampling)
 Chain 3: 0.086 seconds (Total)
 Chain 3:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).

Chain 4:
 Chain 4: Gradient evaluation took 3e-06 seconds
 Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 0.03 seconds.
 Chain 4: Adjust your expectations accordingly!
 Chain 4:
 Chain 4:
 Chain 4: Iteration: 1 / 2000 [0%] (Warmup)
 Chain 4: Iteration: 200 / 2000 [10%] (Warmup)
 Chain 4: Iteration: 400 / 2000 [20%] (Warmup)
 Chain 4: Iteration: 600 / 2000 [30%] (Warmup)
 Chain 4: Iteration: 800 / 2000 [40%] (Warmup)
 Chain 4: Iteration: 1000 / 2000 [50%] (Warmup)
 Chain 4: Iteration: 1001 / 2000 [50%] (Sampling)
 Chain 4: Iteration: 1200 / 2000 [60%] (Sampling)
 Chain 4: Iteration: 1400 / 2000 [70%] (Sampling)
 Chain 4: Iteration: 1600 / 2000 [80%] (Sampling)
 Chain 4: Iteration: 1800 / 2000 [90%] (Sampling)
 Chain 4: Iteration: 2000 / 2000 [100%] (Sampling)
 Chain 4:
 Chain 4: Elapsed Time: 0.046 seconds (Warm-up)
 Chain 4: 0.024 seconds (Sampling)
 Chain 4: 0.07 seconds (Total)
 Chain 4:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).

Chain 1:
 Chain 1: Gradient evaluation took 6e-06 seconds
 Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 0.06 seconds.
 Chain 1: Adjust your expectations accordingly!
 Chain 1:
 Chain 1:
 Chain 1: Iteration: 1 / 2000 [0%] (Warmup)
 Chain 1: Iteration: 200 / 2000 [10%] (Warmup)

```

Chain 1: Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1: Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1: Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1: Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1: Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1: Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1: Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1: Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1: Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1:
Chain 1: Elapsed Time: 0.05 seconds (Warm-up)
Chain 1: 0.028 seconds (Sampling)
Chain 1: 0.078 seconds (Total)
Chain 1:

```

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).

```

Chain 2:
Chain 2: Gradient evaluation took 4e-06 seconds
Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 0.04 seconds.
Chain 2: Adjust your expectations accordingly!
Chain 2:
Chain 2:
Chain 2: Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2: Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2: Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2: Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2: Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2: Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2: Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2: Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2: Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2: Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2: Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2:
Chain 2: Elapsed Time: 0.046 seconds (Warm-up)
Chain 2: 0.029 seconds (Sampling)
Chain 2: 0.075 seconds (Total)
Chain 2:

```

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).

```

Chain 3:

```

Chain 3: Gradient evaluation took 4e-06 seconds
Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 0.04 seconds.
Chain 3: Adjust your expectations accordingly!
Chain 3:
Chain 3:
Chain 3: Iteration: 1 / 2000 [0%] (Warmup)
Chain 3: Iteration: 200 / 2000 [10%] (Warmup)
Chain 3: Iteration: 400 / 2000 [20%] (Warmup)
Chain 3: Iteration: 600 / 2000 [30%] (Warmup)
Chain 3: Iteration: 800 / 2000 [40%] (Warmup)
Chain 3: Iteration: 1000 / 2000 [50%] (Warmup)
Chain 3: Iteration: 1001 / 2000 [50%] (Sampling)
Chain 3: Iteration: 1200 / 2000 [60%] (Sampling)
Chain 3: Iteration: 1400 / 2000 [70%] (Sampling)
Chain 3: Iteration: 1600 / 2000 [80%] (Sampling)
Chain 3: Iteration: 1800 / 2000 [90%] (Sampling)
Chain 3: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3:
Chain 3: Elapsed Time: 0.056 seconds (Warm-up)
Chain 3: 0.027 seconds (Sampling)
Chain 3: 0.083 seconds (Total)
Chain 3:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).

Chain 4:
Chain 4: Gradient evaluation took 5e-06 seconds
Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 0.05 seconds.
Chain 4: Adjust your expectations accordingly!
Chain 4:
Chain 4:
Chain 4: Iteration: 1 / 2000 [0%] (Warmup)
Chain 4: Iteration: 200 / 2000 [10%] (Warmup)
Chain 4: Iteration: 400 / 2000 [20%] (Warmup)
Chain 4: Iteration: 600 / 2000 [30%] (Warmup)
Chain 4: Iteration: 800 / 2000 [40%] (Warmup)
Chain 4: Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4: Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4: Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4: Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4: Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4: Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4:

Chain 4: Elapsed Time: 0.049 seconds (Warm-up)
Chain 4: 0.026 seconds (Sampling)
Chain 4: 0.075 seconds (Total)
Chain 4:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).

Chain 1:
Chain 1: Gradient evaluation took 6e-06 seconds
Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 0.06 seconds.
Chain 1: Adjust your expectations accordingly!
Chain 1:
Chain 1:
Chain 1: Iteration: 1 / 2000 [0%] (Warmup)
Chain 1: Iteration: 200 / 2000 [10%] (Warmup)
Chain 1: Iteration: 400 / 2000 [20%] (Warmup)
Chain 1: Iteration: 600 / 2000 [30%] (Warmup)
Chain 1: Iteration: 800 / 2000 [40%] (Warmup)
Chain 1: Iteration: 1000 / 2000 [50%] (Warmup)
Chain 1: Iteration: 1001 / 2000 [50%] (Sampling)
Chain 1: Iteration: 1200 / 2000 [60%] (Sampling)
Chain 1: Iteration: 1400 / 2000 [70%] (Sampling)
Chain 1: Iteration: 1600 / 2000 [80%] (Sampling)
Chain 1: Iteration: 1800 / 2000 [90%] (Sampling)
Chain 1: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1:
Chain 1: Elapsed Time: 0.052 seconds (Warm-up)
Chain 1: 0.027 seconds (Sampling)
Chain 1: 0.079 seconds (Total)
Chain 1:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).

Chain 2:
Chain 2: Gradient evaluation took 4e-06 seconds
Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 0.04 seconds.
Chain 2: Adjust your expectations accordingly!
Chain 2:
Chain 2:
Chain 2: Iteration: 1 / 2000 [0%] (Warmup)
Chain 2: Iteration: 200 / 2000 [10%] (Warmup)
Chain 2: Iteration: 400 / 2000 [20%] (Warmup)
Chain 2: Iteration: 600 / 2000 [30%] (Warmup)
Chain 2: Iteration: 800 / 2000 [40%] (Warmup)
Chain 2: Iteration: 1000 / 2000 [50%] (Warmup)

Chain 2: Iteration: 1001 / 2000 [50%] (Sampling)
Chain 2: Iteration: 1200 / 2000 [60%] (Sampling)
Chain 2: Iteration: 1400 / 2000 [70%] (Sampling)
Chain 2: Iteration: 1600 / 2000 [80%] (Sampling)
Chain 2: Iteration: 1800 / 2000 [90%] (Sampling)
Chain 2: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2:
Chain 2: Elapsed Time: 0.066 seconds (Warm-up)
Chain 2: 0.028 seconds (Sampling)
Chain 2: 0.094 seconds (Total)
Chain 2:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).

Chain 3:
Chain 3: Gradient evaluation took 4e-06 seconds
Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 0.04 seconds.
Chain 3: Adjust your expectations accordingly!
Chain 3:
Chain 3:
Chain 3: Iteration: 1 / 2000 [0%] (Warmup)
Chain 3: Iteration: 200 / 2000 [10%] (Warmup)
Chain 3: Iteration: 400 / 2000 [20%] (Warmup)
Chain 3: Iteration: 600 / 2000 [30%] (Warmup)
Chain 3: Iteration: 800 / 2000 [40%] (Warmup)
Chain 3: Iteration: 1000 / 2000 [50%] (Warmup)
Chain 3: Iteration: 1001 / 2000 [50%] (Sampling)
Chain 3: Iteration: 1200 / 2000 [60%] (Sampling)
Chain 3: Iteration: 1400 / 2000 [70%] (Sampling)
Chain 3: Iteration: 1600 / 2000 [80%] (Sampling)
Chain 3: Iteration: 1800 / 2000 [90%] (Sampling)
Chain 3: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3:
Chain 3: Elapsed Time: 0.061 seconds (Warm-up)
Chain 3: 0.025 seconds (Sampling)
Chain 3: 0.086 seconds (Total)
Chain 3:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).

Chain 4:
Chain 4: Gradient evaluation took 3e-06 seconds
Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 0.03 seconds.
Chain 4: Adjust your expectations accordingly!
Chain 4:

```

Chain 4:
Chain 4: Iteration:    1 / 2000 [  0%] (Warmup)
Chain 4: Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 4: Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 4: Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 4: Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 4: Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 4: Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 4: Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 4: Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 4: Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 4: Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 4: Iteration:  2000 / 2000 [100%] (Sampling)
Chain 4:
Chain 4: Elapsed Time: 0.061 seconds (Warm-up)
Chain 4:                0.027 seconds (Sampling)
Chain 4:                0.088 seconds (Total)
Chain 4:

```

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).

```

Chain 1:
Chain 1: Gradient evaluation took 6e-06 seconds
Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 0.06 seconds.
Chain 1: Adjust your expectations accordingly!
Chain 1:
Chain 1:
Chain 1: Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1: Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1: Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1: Iteration:   600 / 2000 [ 30%] (Warmup)
Chain 1: Iteration:   800 / 2000 [ 40%] (Warmup)
Chain 1: Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1: Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1: Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1: Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1: Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1: Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1: Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1:
Chain 1: Elapsed Time: 0.056 seconds (Warm-up)
Chain 1:                0.028 seconds (Sampling)
Chain 1:                0.084 seconds (Total)
Chain 1:

```


SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).

Chain 2:

Chain 2: Gradient evaluation took 4e-06 seconds

Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 0.04 seconds.

Chain 2: Adjust your expectations accordingly!

Chain 2:

Chain 2:

Chain 2: Iteration: 1 / 2000 [0%] (Warmup)

Chain 2: Iteration: 200 / 2000 [10%] (Warmup)

Chain 2: Iteration: 400 / 2000 [20%] (Warmup)

Chain 2: Iteration: 600 / 2000 [30%] (Warmup)

Chain 2: Iteration: 800 / 2000 [40%] (Warmup)

Chain 2: Iteration: 1000 / 2000 [50%] (Warmup)

Chain 2: Iteration: 1001 / 2000 [50%] (Sampling)

Chain 2: Iteration: 1200 / 2000 [60%] (Sampling)

Chain 2: Iteration: 1400 / 2000 [70%] (Sampling)

Chain 2: Iteration: 1600 / 2000 [80%] (Sampling)

Chain 2: Iteration: 1800 / 2000 [90%] (Sampling)

Chain 2: Iteration: 2000 / 2000 [100%] (Sampling)

Chain 2:

Chain 2: Elapsed Time: 0.071 seconds (Warm-up)

Chain 2: 0.028 seconds (Sampling)

Chain 2: 0.099 seconds (Total)

Chain 2:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).

Chain 3:

Chain 3: Gradient evaluation took 4e-06 seconds

Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 0.04 seconds.

Chain 3: Adjust your expectations accordingly!

Chain 3:

Chain 3:

Chain 3: Iteration: 1 / 2000 [0%] (Warmup)

Chain 3: Iteration: 200 / 2000 [10%] (Warmup)

Chain 3: Iteration: 400 / 2000 [20%] (Warmup)

Chain 3: Iteration: 600 / 2000 [30%] (Warmup)

Chain 3: Iteration: 800 / 2000 [40%] (Warmup)

Chain 3: Iteration: 1000 / 2000 [50%] (Warmup)

Chain 3: Iteration: 1001 / 2000 [50%] (Sampling)

Chain 3: Iteration: 1200 / 2000 [60%] (Sampling)

Chain 3: Iteration: 1400 / 2000 [70%] (Sampling)

Chain 3: Iteration: 1600 / 2000 [80%] (Sampling)

Chain 3: Iteration: 1800 / 2000 [90%] (Sampling)
Chain 3: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3:
Chain 3: Elapsed Time: 0.048 seconds (Warm-up)
Chain 3: 0.026 seconds (Sampling)
Chain 3: 0.074 seconds (Total)
Chain 3:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).

Chain 4:
Chain 4: Gradient evaluation took 4e-06 seconds
Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 0.04 seconds.
Chain 4: Adjust your expectations accordingly!
Chain 4:
Chain 4:
Chain 4: Iteration: 1 / 2000 [0%] (Warmup)
Chain 4: Iteration: 200 / 2000 [10%] (Warmup)
Chain 4: Iteration: 400 / 2000 [20%] (Warmup)
Chain 4: Iteration: 600 / 2000 [30%] (Warmup)
Chain 4: Iteration: 800 / 2000 [40%] (Warmup)
Chain 4: Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4: Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4: Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4: Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4: Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4: Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4:
Chain 4: Elapsed Time: 0.056 seconds (Warm-up)
Chain 4: 0.027 seconds (Sampling)
Chain 4: 0.083 seconds (Total)
Chain 4:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).

Chain 1:
Chain 1: Gradient evaluation took 5e-06 seconds
Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 0.05 seconds.
Chain 1: Adjust your expectations accordingly!
Chain 1:
Chain 1:
Chain 1: Iteration: 1 / 2000 [0%] (Warmup)
Chain 1: Iteration: 200 / 2000 [10%] (Warmup)
Chain 1: Iteration: 400 / 2000 [20%] (Warmup)

```

Chain 1: Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1: Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1: Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1: Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1: Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 1: Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 1: Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 1: Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1:
Chain 1: Elapsed Time: 0.056 seconds (Warm-up)
Chain 1:                0.026 seconds (Sampling)
Chain 1:                0.082 seconds (Total)
Chain 1:

```

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).

```

Chain 2:
Chain 2: Gradient evaluation took 4e-06 seconds
Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 0.04 seconds.
Chain 2: Adjust your expectations accordingly!
Chain 2:
Chain 2:
Chain 2: Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2: Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2: Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 2: Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 2: Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 2: Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 2: Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 2: Iteration: 1200 / 2000 [ 60%] (Sampling)
Chain 2: Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2: Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2: Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2:
Chain 2: Elapsed Time: 0.045 seconds (Warm-up)
Chain 2:                0.025 seconds (Sampling)
Chain 2:                0.07 seconds (Total)
Chain 2:

```

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).

```

Chain 3:
Chain 3: Gradient evaluation took 3e-06 seconds

```

Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 0.03 seconds.
Chain 3: Adjust your expectations accordingly!
Chain 3:
Chain 3:
Chain 3: Iteration: 1 / 2000 [0%] (Warmup)
Chain 3: Iteration: 200 / 2000 [10%] (Warmup)
Chain 3: Iteration: 400 / 2000 [20%] (Warmup)
Chain 3: Iteration: 600 / 2000 [30%] (Warmup)
Chain 3: Iteration: 800 / 2000 [40%] (Warmup)
Chain 3: Iteration: 1000 / 2000 [50%] (Warmup)
Chain 3: Iteration: 1001 / 2000 [50%] (Sampling)
Chain 3: Iteration: 1200 / 2000 [60%] (Sampling)
Chain 3: Iteration: 1400 / 2000 [70%] (Sampling)
Chain 3: Iteration: 1600 / 2000 [80%] (Sampling)
Chain 3: Iteration: 1800 / 2000 [90%] (Sampling)
Chain 3: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3:
Chain 3: Elapsed Time: 0.063 seconds (Warm-up)
Chain 3: 0.028 seconds (Sampling)
Chain 3: 0.091 seconds (Total)
Chain 3:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).

Chain 4:
Chain 4: Gradient evaluation took 3e-06 seconds
Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 0.03 seconds.
Chain 4: Adjust your expectations accordingly!
Chain 4:
Chain 4:
Chain 4: Iteration: 1 / 2000 [0%] (Warmup)
Chain 4: Iteration: 200 / 2000 [10%] (Warmup)
Chain 4: Iteration: 400 / 2000 [20%] (Warmup)
Chain 4: Iteration: 600 / 2000 [30%] (Warmup)
Chain 4: Iteration: 800 / 2000 [40%] (Warmup)
Chain 4: Iteration: 1000 / 2000 [50%] (Warmup)
Chain 4: Iteration: 1001 / 2000 [50%] (Sampling)
Chain 4: Iteration: 1200 / 2000 [60%] (Sampling)
Chain 4: Iteration: 1400 / 2000 [70%] (Sampling)
Chain 4: Iteration: 1600 / 2000 [80%] (Sampling)
Chain 4: Iteration: 1800 / 2000 [90%] (Sampling)
Chain 4: Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4:
Chain 4: Elapsed Time: 0.048 seconds (Warm-up)

```
Chain 4:          0.027 seconds (Sampling)
Chain 4:          0.075 seconds (Total)
Chain 4:
```

```
Fitting imputed model 1
```

```
Start sampling
```

```
Fitting imputed model 2
```

```
Start sampling
```

```
Fitting imputed model 3
```

```
Start sampling
```

```
Fitting imputed model 4
```

```
Start sampling
```

```
Fitting imputed model 5
```

```
Start sampling
```

```
summary(brms_fitted)
```

```
Family: MV(gaussian, gaussian)
Links: mu = identity
       mu = identity
Formula: final ~ hwrte + ACT
         hwrte ~ ACT
Data: imputed_data (Number of observations: 680)
Draws: 20 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 20000
```

```
Regression Coefficients:
```

	Estimate	Est.Error	1-95% CI	u-95% CI
final_Intercept	10.50	1.36	7.85	13.17
hwrte_Intercept	101.47	4.93	91.92	111.07

final_hwrte	0.04	0.01	0.03	0.06
final_ACT	0.51	0.05	0.42	0.61
hwrte_ACT	-0.61	0.22	-1.04	-0.19

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI
sigma_final	4.32	0.12	4.09	4.56
sigma_hwrte	19.60	0.54	18.57	20.70

Draws were sampled using sampling(NUTS). Overall Rhat and ESS estimates are not informative for brm_multiple models and are hence not displayed. Please see ?brm_multiple for how to assess convergence of such models.

```
library(bayestestR) # a package that facilitates mediation analysis with brms
mediation(brms_fitted)
```

Causal Mediation Analysis for Stan Model

```
Treatment: ACT
Mediator : hwrte
Response : final
```

Effect	Estimate	95% ETI
Direct Effect (ADE)	0.514	[0.421, 0.607]
Indirect Effect (ACME)	-0.026	[-0.050, -0.008]
Mediator Effect	0.044	[0.027, 0.060]
Total Effect	0.488	[0.393, 0.581]

Proportion mediated: -5.31% [-10.05%, -0.56%]

Direct and indirect effects have opposite directions. The proportion mediated is not meaningful.